

AgriAdviser-NG: A Domain-Specific Small Language Model for Climate-Aware Rice & Maize Advisory to Smallholder Farmers in Nigeria

Problem Statement

1. Background and Context

More than 80% of Nigeria's farmers are smallholders, and they produce most of the country's cereals, legumes, and other staple crops. Rice and maize sit at the center of this, both as household food staples and as cash crops, which makes the quality of the agronomic advice these farmers can access a direct driver of national food security and rural income.

The advice they need is concrete and localized: which variety and seed rate suits their zone, how much and when to apply fertilizer, which herbicide or pest control step applies to their crop stage, and how shifting rainfall patterns should change the timing of these decisions. That information already exists, held in state-level Agricultural Development Programme (ADP) fertilizer tables, IITA/AfricaRice variety bulletins, NAERLS extension manuals, and NiMet rainfall bulletins, but it is scattered across technical documents that farmers cannot search or interpret themselves.

The channel meant to bridge this gap, Nigeria's agricultural extension system, is severely under-resourced relative to the population it serves. The recommended ratio is one extension officer for every 800 to 1,000 farming households, but the national average reached one officer for every 6,446 households in 2024, a figure that hides sharp regional disparities: Akwa Ibom, for example, has one extension officer per 24,468 farming households. Industry stakeholders warn this ratio could worsen further as the current cohort of extension workers, largely hired in the 1990s, retires without adequate replacement. Much of the advisory content that does exist is delivered through TV and radio, channels many rural farmers cannot rely on, and even when a farmer does reach an officer, the answer they need still has to be found in the right document.

2. Core Problem

- A critical deficit of real-time, localized agronomic intelligence: farmers lack a fast way to get a site-specific answer (correct rate, timing, and variety for their crop, state, and season) rather than generic guidance.
- Growing vulnerability to sudden climate deviations and pest or disease pressure, such as erratic rainfall onset, fall armyworm in maize, and rice blast, where a delayed or generic recommendation directly reduces yield.
- Infrastructural barriers, most rural farmers and many extension workers operate without stable internet or reliable electricity, ruling out any solution that assumes persistent connectivity or high-end hardware.
- A need for resource-efficient recommendations, since fertilizer, seed, and water are costly inputs for smallholders, imprecise advice does not just under-inform, it wastes money and inputs at farm scale.

3. Current Approach and Limitations

Where digital advisory has been tried, the default approach has been to point farmers toward large-scale, cloud-based Large Language Models (LLMs) such as ChatGPT. A 2024 study evaluating ChatGPT against real farmer questions on irrigated lowland rice cultivation in Nigeria confirmed this is a solvable NLP problem, not just a policy one, but it also identified a specific failure mode worth designing around: general-purpose LLMs lack the site-specific, agronomic precision (correct input rates and timing for local conditions) that farmers actually need, and instead default to global averages.

Beyond that accuracy gap, cloud-based LLMs carry limitations that make them a poor fit for this population specifically:

- Prohibitive computational and operational cost for sustained deployment to a smallholder population at scale.
- Incompatibility with offline use, these models assume persistent, high-bandwidth internet access that most rural farmers and extension workers do not have.
- Latency and text-fragmentation issues during tokenization, general-purpose tokenizers were not built for Nigerian English, Pidgin, or code-mixed farmer phrasing, which fragments queries and slows responses.
- Weak capture of domain-specific terminology and local agronomic phrasing, general corpora underrepresent Nigerian crop, input, and variety names.
- Hardware requirements that exceed the low-tier feature phones most extension workers and farmers actually carry.

4. Research Aim

The main aim of this project is to develop and evaluate a document-retrieval approach that improves the ranking of relevant agricultural extension documents against smallholder farmers' rice and maize questions, so that a small, resource-efficient language model can be paired with the correct site-specific fact rather than relying on its own (unreliable) parametric memory.

Specifically, the research will:

- Test embedding-based and hybrid retrieval methods for matching a farmer's natural-language question to the correct crop, state, season, and variety-specific fact drawn from ADP fertilizer tables, IITA/AfricaRice bulletins, NAERLS manuals, and NiMet bulletins.
- Compare these methods against a TF-IDF retrieval baseline.
- Target a measurable improvement in ranking quality over that baseline, with the explicit goal of reducing the inaccurate, site-specific answers that the 2024 ChatGPT study identified as the core failure mode.

5. Proposed Technical Approach

Domain Adaptation & Tokenization

- QLoRA fine-tuning of a genuinely small open model (Phi or Gemma) on localized agronomic corpora, ADP fertilizer tables, IITA/AfricaRice variety bulletins, NAERLS extension manuals, and NiMet rainfall bulletins, so the model learns domain phrasing and intent rather than memorizing facts it cannot reliably retain at this scale.
- Tokenizer handling tuned for Nigerian English and code-mixed Pidgin farmer phrasing, so common local terms for crops, inputs, and pests are not fragmented into meaningless subwords.

Edge Optimization

- GGUF quantization of the fine-tuned model for CPU-only inference, matching the 2–3GB VRAM footprint needed to run on a single consumer GPU or ordinary edge hardware.
- An offline-first inference and retrieval path, so the system keeps working through the intermittent connectivity and electricity common in rural deployment areas.

Retrieval Layer

- A lightweight retrieval database holding site-specific facts (rate, timing, variety, soil zone) separately from the model's weights, with the model's role limited to identifying the crop/state/season the farmer is asking about, retrieving the matching fact, and phrasing a fluent answer around it.
- An explicit abstention path: when retrieval finds no matching fact, the model is trained to say so rather than guess.

Satellite-derived remote sensing and soil-API integration (e.g., Sentinel imagery, iSDAsoil metadata) are deliberately excluded from this phase, they add real value but also add connectivity and cost assumptions that conflict with the feature-phone-first design goal; they are noted as a future extension in Section 6.

6. Scope and Boundaries

In scope

- Rice and maize advisory only, covering the crop/state/season/variety combinations represented in the ADP, IITA/AfricaRice, NAERLS, and NiMet source documents.
- Text-based question answering delivered over USSD and SMS, with WhatsApp as a secondary channel, sized to feature-phone constraints.
- Farmer queries in Nigerian English and common Pidgin phrasing.

Out of scope

- Crops other than rice and maize, and states or seasons not represented in the current source documents.
- Multi-modal inputs, satellite/remote-sensing data, soil-sensor feeds, or image-based pest and disease diagnosis.
- Voice interfaces, and full replacement of human extension officers rather than augmentation of their reach.

7. Evaluation Method

- Retrieval quality will be measured using nDCG@5 on a held-out set of representative farmer questions.
- Results will be compared directly against a TF-IDF retrieval baseline run on the same question set and document store.
- Where possible, this will be paired with a secondary check on the abstention path, confirming the model declines to answer when retrieval returns no matching fact, rather than guessing.

8. Significance

This project targets the extension gap directly rather than around it: instead of assuming more extension officers or more bandwidth, it makes existing agronomic documentation searchable and answerable at the point of need, on the feature phones and intermittent networks farmers already have. By separating retrieval from generation, it also directly addresses the specific failure mode the 2024 ChatGPT study surfaced, inaccurate, non-localized answers to site-specific farming questions, while keeping compute and cost low enough to be viable for Nigeria's smallholder population at scale. If the retrieval-plus-small-model pattern validated here performs well against the TF-IDF baseline, it offers a low-cost template that could extend beyond rice and maize to other crops and regions facing the same extension shortfall.